AP PRECALCULUS

Unit 1 • Practice • Answer key

Polynomial and Rational Functions

AP-style practice • Multiple choice and free response • Answer key with explanations

About this practice set

The questions follow the format of the AP Precalculus exam. Part A and Part B are multiple choice; Part C is free response. Each question is labeled **NO CALCULATOR** or **CALCULATOR**. Suggested timing: Part A 35 min • Part B 10 min • Part C 45 min.

Part A • Multiple choice • No calculator

Directions • Part A

A calculator may **not** be used on this part. Solve each problem, then choose the best of the four answer choices. Unless otherwise stated, the domain of a function f is the set of all real numbers x for which $f(x)$ is a real number. Some questions refer to a graph: read the values you need from the grid.

Question 1 (1.1)

NO CALCULATOR

The table gives values of a function f at selected values of x.

x	0	2	4	6
$f(x)$	5	9	11	12

Which statement is consistent with the table?

(A) f is increasing and its graph is concave up.

(B) f is increasing and its graph is concave down.

(C) f is decreasing and its graph is concave up.

(D) f is decreasing and its graph is concave down.

Solution

Answer: (B). The outputs increase, so f is increasing. The average rates of change on $[0,2]$, $[2,4]$, $[4,6]$ are 2, 1, 0.5: the rate of change is decreasing, so the graph is concave down.

Question 2 (1.2)

NO CALCULATOR

What is the average rate of change of $f(x) = x^3 - 2x$ on the interval $[-1, 2]$?

(A) 3

(B) $\dfrac{5}{3}$

(C) 1

(D) -1

Solution

Answer: (C). $f(2) = 8 - 4 = 4$ and $f(-1) = -1 + 2 = 1$, so $\dfrac{4-1}{2-(-1)} = \dfrac{3}{3} = 1$.

Question 3 (1.3)

NO CALCULATOR

x	1	2	3	4	5
y	2	5	10	17	26

Which type of function best models the data in the table?

(A) Quadratic
(B) Linear
(C) Cubic
(D) Constant

Solution
Answer: (A). The first differences are $3, 5, 7, 9$ and the second differences are all 2. Constant second differences over equal input intervals indicate a quadratic model ($y = x^2 + 1$).

Question 4 (1.5)

NO CALCULATOR

A polynomial p of degree 4 has real coefficients. Two of its zeros are 2 and $1 + 3i$. Which of the following must also be a zero of p?

(A) -2
(B) $3 + i$
(C) $-1 - 3i$
(D) $1 - 3i$

Solution
Answer: (D). Non-real zeros of a polynomial with real coefficients come in conjugate pairs, so $1 - 3i$ is a zero.

Question 5 (1.6)

NO CALCULATOR

Let $p(x) = x^2(x-3)^3(x+1)$. What is $\lim_{x \to -\infty} p(x)$?

(A) $-\infty$
(B) $+\infty$
(C) 0
(D) 1

Solution
Answer: (B). The leading term is $x^2 \cdot x^3 \cdot x = x^6$: even degree with positive leading coefficient, so $p(x) \to +\infty$ as $x \to -\infty$.

Question 6 (1.5)

NO CALCULATOR

Which statement describes the graph of $p(x) = (x+2)^2(x-1)$ at $x = -2$?

(A) It has a vertical asymptote.
(B) It crosses the x-axis.
(C) It touches the x-axis without crossing it.
(D) It has a hole.

Solution
Answer: (C). $x = -2$ is a zero of even multiplicity (2), so the graph touches the x-axis and turns back. Polynomials have no asymptotes or holes.

Question 7 (1.7) NO CALCULATOR

Which line is a horizontal asymptote of the graph of $r(x) = \dfrac{3x^2 - x}{6x^2 + 5}$?

(A) $y = 3$
(B) $y = 0$
(C) $y = -\dfrac{1}{5}$
(D) $y = \dfrac{1}{2}$

Solution
Answer: (D). The degrees are equal, so the asymptote is the ratio of leading coefficients: $y = \frac{3}{6} = \frac{1}{2}$.

Question 8 (1.10) NO CALCULATOR

Let $r(x) = \dfrac{x^2 - 1}{x^2 + x - 2}$. Which statement is true about the graph of r at $x = 1$?

(A) It has a vertical asymptote $x = 1$.
(B) It has a hole at $\left(1, \frac{2}{3}\right)$.
(C) It has a zero at $x = 1$.
(D) It has a hole at $(1, 0)$.

Solution
Answer: (B). $r(x) = \dfrac{(x-1)(x+1)}{(x-1)(x+2)}$. The factor $(x-1)$ cancels, so there is a hole at $x = 1$. Its height is $\dfrac{1+1}{1+2} = \dfrac{2}{3}$.

Question 9 (1.9) NO CALCULATOR

What is $\lim\limits_{x \to 2^+} \dfrac{x+1}{2-x}$?

(A) 0
(B) 3
(C) $+\infty$
(D) $-\infty$

Solution
Answer: (D). As $x \to 2^+$, the numerator approaches $3 > 0$ and the denominator $2 - x$ approaches 0 through negative values. The quotient decreases without bound.

Question 10 (1.11) NO CALCULATOR

The graph of $r(x) = \dfrac{2x^2 + 3x - 4}{x + 2}$ has a slant asymptote. What is its equation?

(A) $y = 2x - 1$
(B) $y = 2x + 7$
(C) $y = 2x + 3$
(D) $y = x - 1$

Solution
Answer: (A). Synthetic division by -2 on $2,\ 3,\ -4$ gives $2,\ -1$ with remainder -2, so $r(x) = 2x - 1 - \dfrac{2}{x+2}$. As $x \to \pm\infty$ the fraction goes to 0.

Question 11 (1.11)

NO CALCULATOR

What is the coefficient of x^2 in the expansion of $(x-3)^4$?

(A) 6
(B) -108
(C) 54
(D) 9

Solution
Answer: (C). Row 4 of Pascal's triangle is $1, 4, 6, 4, 1$. The x^2 term is $6\,x^2(-3)^2 = 54x^2$.

Question 12 (1.12)

NO CALCULATOR

The function f has domain $[0, 8]$ and range $[-2, 3]$. What is the range of $g(x) = -2f\left(\frac{x}{2}\right) + 1$?

(A) $[-5, 5]$
(B) $[-3, 7]$
(C) $[-4, 6]$
(D) $[-1, 3]$

Solution
Answer: (A). The horizontal dilation does not change the range. Multiplying $[-2, 3]$ by -2 gives $[-6, 4]$; adding 1 gives $[-5, 5]$.

Question 13 (1.5)

NO CALCULATOR

Which function is even?

(A) $f(x) = x^3 - x$
(B) $f(x) = x^4 + 3x^2 - 1$
(C) $f(x) = x^2 + x$
(D) $f(x) = (x-1)^2$

Solution
Answer: (B). $(-x)^4 + 3(-x)^2 - 1 = x^4 + 3x^2 - 1$, so $f(-x) = f(x)$. The others fail this test (for example, (C) gives $x^2 - x$).

Question 14 (1.8)

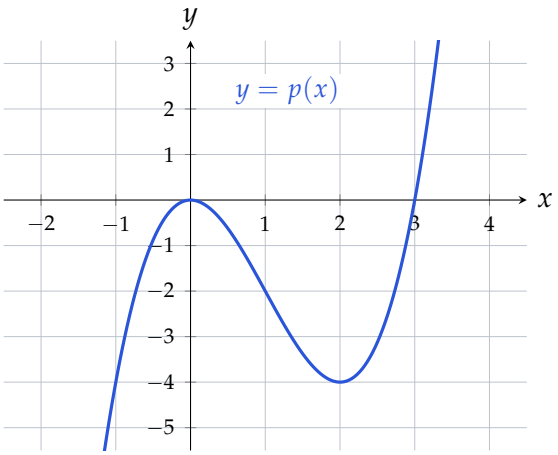
NO CALCULATOR

What is the solution set of $\dfrac{x-3}{x+1} \geq 0$?

(A) $(-\infty, -1] \cup [3, \infty)$
(B) $[3, \infty)$
(C) $(-1, 3]$
(D) $(-\infty, -1) \cup [3, \infty)$

Solution
Answer: (D). The sign can change only at the zero $x = 3$ and at $x = -1$ (excluded). Test points: $x = -2$ gives $\frac{-5}{-1} > 0$, $x = 0$ gives $-3 < 0$, $x = 4$ gives $\frac{1}{5} > 0$. Include $x = 3$ (value 0) but not $x = -1$.

Questions 15 to 18 refer to the graph of the polynomial p below.



Question 15 (1.5 • graph) NO CALCULATOR

Which of the following could be an expression for $p(x)$?

(A) $x(x-3)^2$
(B) $-x^2(x-3)$
(C) $x^2(x-3)$
(D) $x^2(x+3)$

Solution
Answer: (C). The graph touches the x-axis at $x = 0$ (even multiplicity) and crosses at $x = 3$ (odd multiplicity), and it rises to the right, so the leading coefficient is positive. Check: $p(2) = 4(-1) = -4$ matches the graph.

Question 16 (1.4 • graph) NO CALCULATOR

On which interval is p decreasing?

(A) $(-\infty, 0)$
(B) $(0, 2)$
(C) $(2, \infty)$
(D) $(0, 3)$

Solution
Answer: (B). The graph falls from the local maximum $(0, 0)$ to the local minimum $(2, -4)$ and rises elsewhere.

Question 17 (1.2 • graph) NO CALCULATOR

What is the average rate of change of p on the interval $[1, 3]$?

(A) -2
(B) 0
(C) 2
(D) 1

Solution
Answer: (D). From the graph, $p(1) = -2$ and $p(3) = 0$, so $\dfrac{0-(-2)}{3-1} = 1$.

Question 18 (1.4 • graph) NO CALCULATOR

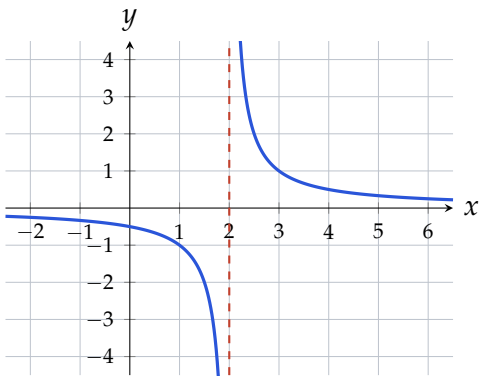
The graph of p has a point of inflection at $(1, -2)$. On which interval is the graph of p concave up?

(A) $(1, \infty)$
(B) $(-\infty, 1)$
(C) $(0, 2)$
(D) $(-\infty, 0)$

Solution
Answer: (A). Left of $x = 1$ the rate of change decreases (the graph bends downward); right of $x = 1$ it increases, so the graph is concave up on $(1, \infty)$.

Question 19 (1.9 • graph) NO CALCULATOR

The graph of a rational function s is shown. It has a vertical asymptote $x = 2$ and a horizontal asymptote $y = 0$, and it passes through $(3, 1)$ and $(1, -1)$. Which could be an expression for $s(x)$?

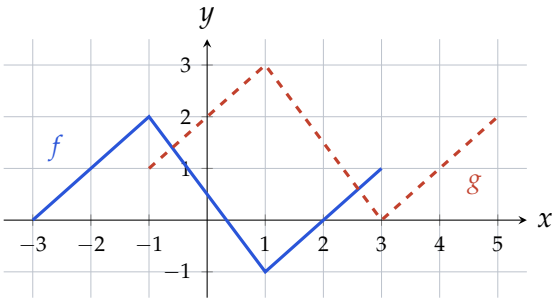


(A) $\dfrac{1}{x+2}$
(B) $\dfrac{1}{x-2}$
(C) $\dfrac{x}{x-2}$
(D) $\dfrac{x-2}{x}$

Solution
Answer: (B). A vertical asymptote at $x = 2$ needs the factor $x - 2$ in the denominator only. $y = 0$ as horizontal asymptote rules out (C) and (D) (their asymptote is $y = 1$). Check: $\frac{1}{3-2} = 1$ and $\frac{1}{1-2} = -1$.

Question 20 (1.12 • graph) NO CALCULATOR

The graphs of f (solid) and g (dashed) are shown. Which is an equation for g in terms of f?



(A) $g(x) = f(x-2)+1$
(B) $g(x) = f(x+2)+1$
(C) $g(x) = f(x-2)-1$
(D) $g(x) = f(x+2)-1$

Solution
Answer: (A). Each point of f moves 2 units right and 1 unit up: $(-3,0) \to (-1,1)$, $(-1,2) \to (1,3)$, $(1,-1) \to (3,0)$. That is $g(x) = f(x-2)+1$.

Part B • Multiple choice • Graphing calculator

Directions • Part B

A graphing calculator is **required** for the questions in this part. If the exact value of the correct answer is not one of the choices, choose the one that best approximates it.

Question 21 (1.14) CALCULATOR

An open box is made from a 16 in by 10 in sheet by cutting squares of side x in from the corners. Its volume is $V(x) = x(16-2x)(10-2x)$ for $0 < x < 5$. What is the maximum volume of the box?

(A) 120 in^3
(B) 144 in^3
(C) 160 in^3
(D) 168 in^3

Solution
Answer: (B). Graph V on $0 < x < 5$ and use the maximum feature: the maximum occurs at $x = 2$, with $V(2) = 2 \cdot 12 \cdot 6 = 144$ in^3.

Question 22 (1.14) CALCULATOR

The height h, in meters, of a model rocket was measured at times t, in seconds.

t	0	1	2	3	4
$h(t)$	1.2	6.1	9.0	9.9	8.6

Using a quadratic regression, what is the predicted height at $t = 5$ seconds?

(A) 4.2 m
(B) 5.3 m
(C) 6.4 m
(D) 7.3 m

Solution
Answer: (B). The quadratic regression is $h(t) \approx -1.029t^2 + 5.974t + 1.183$, so $h(5) \approx 5.34$ m.

Part C • Free response

Directions • Part C

Show all of your work: your answers are graded on the correctness and completeness of your methods as well as on the final answers. When a question asks you to justify or explain, give mathematical reasons in words. Use a calculator only on the question labeled **CALCULATOR**. Decimal answers should be correct to three places after the decimal point.

Question 23 (Function concepts) NO CALCULATOR

The polynomial function h has degree 3 and leading coefficient 1. Selected values of h are given in the table.

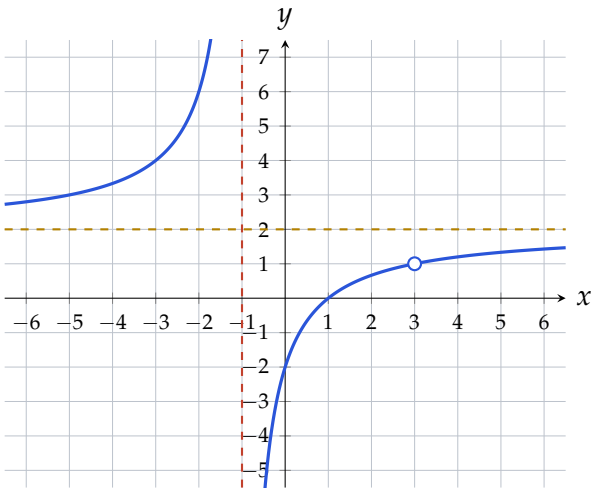
x	-2	-1	0	1	2	3
$h(x)$	-1	3	1	-1	3	19

(A) (i) Find the average rate of change of h over the interval $[-1, 1]$. Show the work that leads to your answer.
(ii) How many real zeros does h have? Give a reason for your answer using the table.
(B) (i) The average rates of change of h over $[1, 2]$ and $[2, 3]$ are different. Use them to determine whether the graph of h is concave up or concave down on $[1, 3]$. Give a reason for your answer.
(ii) Write a limit statement that describes the end behavior of h as x decreases without bound.
(C) Let $k(x) = h(x - 2) + 4$. Find $k(3)$, and describe how the graph of k is obtained from the graph of h.

Solution
(A)(i) $\dfrac{h(1) - h(-1)}{1 - (-1)} = \dfrac{-1 - 3}{2} = -2.$
(ii) h is a polynomial, so it is continuous. Its values change sign on $[-2, -1]$ (from -1 to 3), on $[0, 1]$ (from 1 to -1) and on $[1, 2]$ (from -1 to 3), so h has a zero in each of these intervals: at least 3 real zeros. A degree-3 polynomial has at most 3 zeros, so h has **exactly 3** real zeros.
(B)(i) On $[1, 2]$: $\frac{3-(-1)}{1} = 4$. On $[2, 3]$: $\frac{19-3}{1} = 16$. The average rates of change over consecutive equal-length intervals are increasing, so the graph of h is concave up on $[1, 3]$.
(ii) The leading term is x^3 (odd degree, positive coefficient), so $\lim_{x \to -\infty} h(x) = -\infty$.
(C) $k(3) = h(1) + 4 = -1 + 4 = 3$. The graph of k is the graph of h translated 2 units to the right and 4 units up.

Question 24 (Graph analysis) NO CALCULATOR

The graph of the rational function $r(x) = \dfrac{ax^2+bx+6}{x^2-2x-3}$, where a and b are constants, is shown. The graph has a vertical asymptote $x=-1$, a horizontal asymptote $y=2$, an x-intercept at $(1,0)$, and a hole at $(3,1)$.



(A) (i) Write limit statements that describe the behavior of r near $x=-1$, using the graph.
(ii) Write a limit statement that describes the end behavior of r.

(B) Use the horizontal asymptote and the x-intercept to find the values of a and b. Show that your expression has a hole at $(3,1)$.

(C) Using the graph, solve $r(x)<0$. Express your answer in interval notation.

Solution

(A)(i) $\lim\limits_{x\to -1^-} r(x) = +\infty$ and $\lim\limits_{x\to -1^+} r(x) = -\infty$.

(ii) $\lim\limits_{x\to \pm\infty} r(x) = 2$.

(B) Equal degrees, so the horizontal asymptote is $y=\frac{a}{1}$: $a=2$. Since $r(1)=0$, the numerator is 0 at $x=1$: $2+b+6=0$, so $b=-8$. Then

$$r(x) = \frac{2x^2-8x+6}{x^2-2x-3} = \frac{2(x-1)(x-3)}{(x+1)(x-3)}.$$

The factor $(x-3)$ cancels, so there is a hole at $x=3$, with height $\dfrac{2(3-1)}{3+1} = 1$: the hole is $(3,1)$.

(C) The graph is below the x-axis between the asymptote $x=-1$ and the zero $x=1$: $(-1,1)$.

Question 25 (Modeling a non-periodic context) CALCULATOR

A theater studies how its revenue R, in dollars, depends on the ticket price p, in dollars.

p (dollars)	10	15	20	25	30
R (dollars)	4000	5250	6000	6250	6000

(A) (i) Explain why a quadratic function is an appropriate model for these data.
(ii) Write an expression for a quadratic function $R(p)$ that fits the data exactly.

(B) (i) Use your model to predict the revenue when the ticket price is \$35.
(ii) Find the average rate of change of R from $p=25$ to $p=35$, and interpret its meaning in context, with units.

(C) Revenue cannot be negative. Find the interval of ticket prices for which the model gives a revenue of at least \$0, and state one reason the model may not be valid for every price in that interval.

Solution

(A)(i) The inputs are equally spaced (every \$5). The first differences are $1250, 750, 250, -250$ and the second differences are all -500. Constant second differences indicate a quadratic model.
(ii) The table is symmetric about $p = 25$ ($R(20) = R(30) = 6000$), so the vertex is $(25, 6250)$: $R(p) = a(p-25)^2 + 6250$. Using $R(10) = 4000$: $225a + 6250 = 4000$, so $a = -10$.

$$R(p) = -10(p-25)^2 + 6250.$$

(B)(i) $R(35) = -10(10)^2 + 6250 = 5250$ dollars.
(ii) $\dfrac{R(35) - R(25)}{35 - 25} = \dfrac{5250 - 6250}{10} = -100$. Between a price of \$25 and \$35, revenue decreases by \$100 on average for each \$1 increase in the ticket price.
(C) $R(p) \geq 0 \iff (p-25)^2 \leq 625 \iff 0 \leq p \leq 50$. Possible reasons: the model is built from prices between \$10 and \$30 only; at very high prices attendance may drop to zero sooner than the model predicts; the theater's capacity limits attendance at low prices.

Question 26 (Symbolic manipulations) NO CALCULATOR

Let $r(x) = \dfrac{2x^2 + 2x - 4}{x^2 - 1}$ and $p(x) = x^3 - 2x^2 - 5x + 6$.

(A) (i) Find the zeros of r and the coordinates of any hole in the graph of r.
(ii) Find the vertical asymptote of the graph of r. Write limit statements that justify your answer.
(B) (i) Write a limit statement that describes the end behavior of r.
(ii) Solve $r(x) \geq 0$. Express your answer in interval notation.
(C) It is known that $p(1) = 0$. Write $p(x)$ as a product of linear factors, and find all zeros of p.

Solution

(A)(i) $r(x) = \dfrac{2(x+2)(x-1)}{(x-1)(x+1)}$. The factor $(x-1)$ cancels: hole at $x = 1$, with height $\dfrac{2(1+2)}{1+1} = 3$, so the hole is $(1, 3)$. The only zero is $x = -2$ (note $x = 1$ is not in the domain).
(ii) $x = -1$ is a zero of the denominator that does not cancel. For $x \neq 1$, $r(x) = \dfrac{2(x+2)}{x+1}$. Near $x = -1$ the numerator is close to $2 > 0$, so $\lim\limits_{x \to -1^-} r(x) = -\infty$ and $\lim\limits_{x \to -1^+} r(x) = +\infty$. The line $x = -1$ is a vertical asymptote.
(B)(i) Equal degrees, leading coefficients 2 and 1: $\lim\limits_{x \to \pm\infty} r(x) = 2$.
(ii) Using $\dfrac{2(x+2)}{x+1}$: the sign changes at $x = -2$ and $x = -1$. It is positive on $(-\infty, -2)$, negative on $(-2, -1)$, positive on $(-1, \infty)$. Include $x = -2$; exclude $x = -1$ and the hole $x = 1$:

$$(-\infty, -2] \cup (-1, 1) \cup (1, \infty).$$

(C) Since $p(1) = 0$, $(x-1)$ is a factor. Synthetic division by 1 on $1, -2, -5, 6$ gives $1, -1, -6$ with remainder 0, so $p(x) = (x-1)(x^2 - x - 6) = (x-1)(x-3)(x+2)$. The zeros are $1, 3$ and -2.